

Homework 7

Due October 28, 2025.

Problem 1.

Construct the adjoint representation of $SU(2)$ (antihermitian 3×3 matrices) in two ways:

- a) Using the structure constants $f_{ab}{}^c = \epsilon_{abc}$, and
- b) Using raising (J_+) and lowering (J_-) operators. (In this case J_+ , J_- and J_3 generate $SU(1,1)$ which is the Cartan-Weyl basis for $SU(2)$. So you have to transform from hermitian J_+ and J_- to antihermitian T_1 and T_2 .)

Find the similarity transformation S which maps $T_a^{(II)}$ obtained in the second case to the $T_a^{(I)}$ of the first case

$$ST_a^{(II)}S^{-1} = T_a^{(I)}. \quad (1)$$

Problem 2.

Use the “master formula” of chapter 23 of the notes to construct the representation of $SU(3)$ whose highest weight vector is twice the first fundamental weight vector. Draw the weight diagram. Then draw in this diagram the complex conjugate representation. What is the highest weight of the latter?